

Support Vector Regression

The linear regression model

$$f(x) = x^T \beta + \beta_0$$

To estimate β , we consider minimization of

$$H(\beta, \beta_0) = \sum_{i=1}^N V(y_i - f(x_i)) + \frac{\lambda}{2} \|\beta\|^2$$

with a loss function V and a regularization $\frac{\lambda}{2} \|\beta\|^2$

- How to apply SVM to solve the linear regression problem?

Linear Regression Model (Cont)

The basic idea:

- Given training data set $(x_1, y_1), \dots, (x_N, y_N)$
- Target: find a function $f(x)$ that has at most ϵ deviation from targets y_i for all the training data and at the same time is as less complex (flat) as possible.

In other words we do not care about errors as long as they are less than ϵ but will not accept any deviation larger than this.

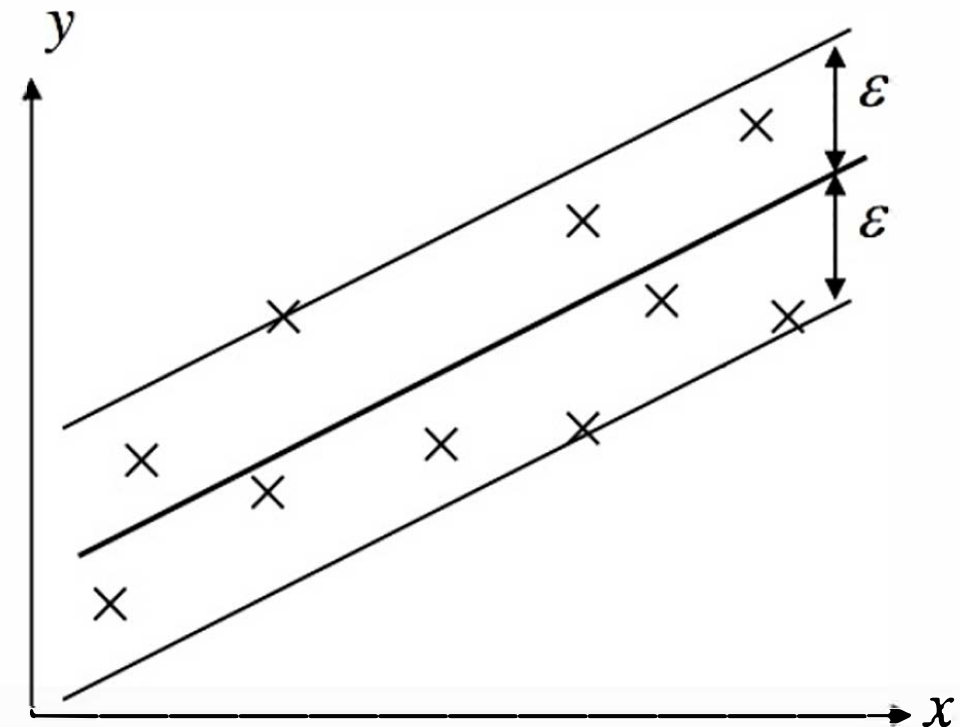
- We want to find one " ϵ -tube" that contains all the samples.
- Intuitively, a tube, with a small width, seems to over-fit with the training data.

We should find $f(x)$ that its ϵ -tube's width is as big as possible (more generalization capability, less prediction error in future).

- With a defined ϵ , a bigger tube corresponds to a smaller β (flatter function).

- Optimization problem:

$$\begin{aligned} & \text{minimize } \frac{1}{2} \|\beta\|^2 \\ & \text{s.t } \begin{cases} y_i - f(x_i) \leq \epsilon \\ f(x_i) - y_i \leq \epsilon \end{cases} \end{aligned}$$

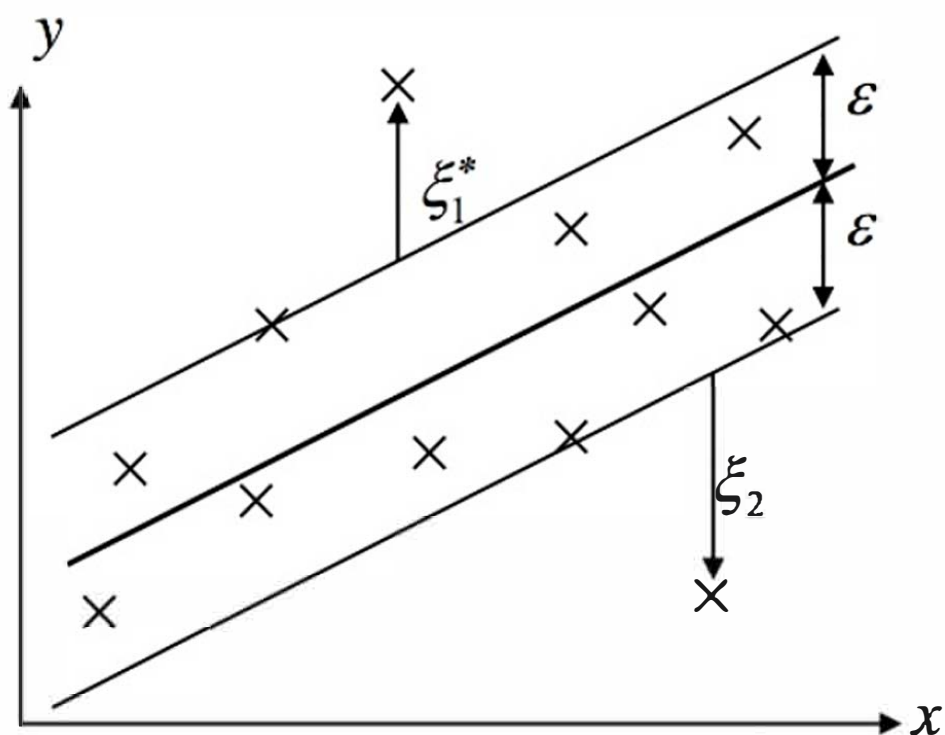


With a defined ϵ , this problem is not always feasible, so we also want to allow some errors.

Use slack variables ξ_i, ξ_i^* , the new optimization problem:

$$\text{minimize } \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^N (\xi_i + \xi_i^*)$$

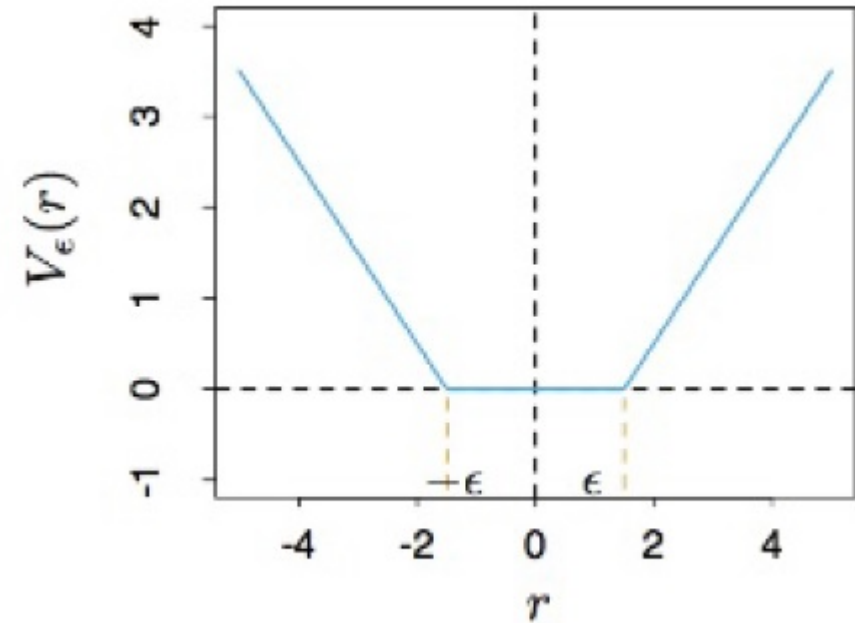
$$\text{s.t } \begin{cases} y_i - f(x_i) \leq \epsilon + \xi_i^* \\ f(x_i) - y_i \leq \epsilon + \xi_i \\ \xi_i, \xi_i^* \geq 0 \end{cases}$$



Let $\lambda = \frac{1}{C}$

Use an " ϵ -insensitive" error measure, ignoring errors of size less than ϵ

$$V_{\epsilon}(r) = \begin{cases} 0 & \text{if } |r| < \epsilon \\ |r| - \epsilon, & \text{otherwise.} \end{cases}$$



We have the minimization of

$$H(\beta, \beta_0) = \sum_{i=1}^N V(y_i - f(x_i)) + \frac{\lambda}{2} \|\beta\|^2$$

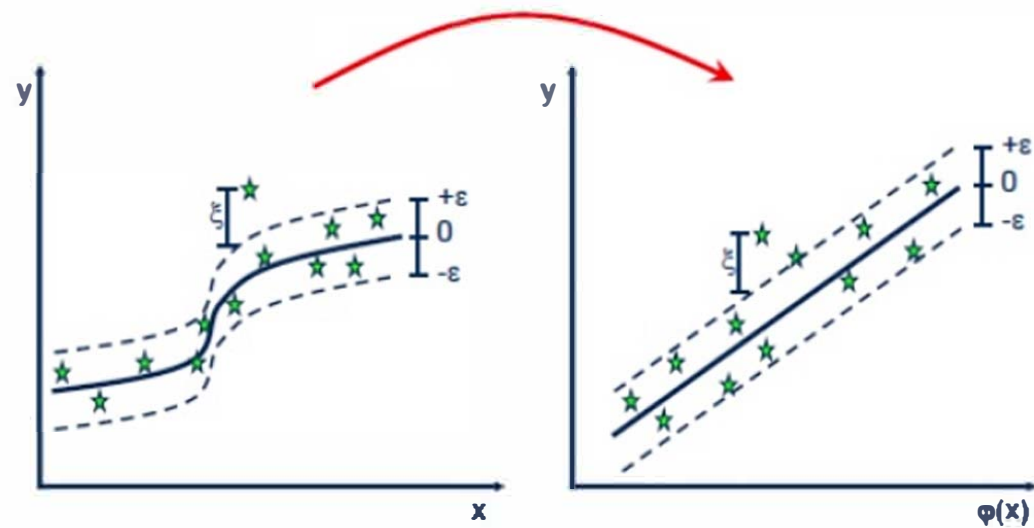
- Parameter ϵ controls the width of the ϵ -insensitive tube. The value of ϵ can affect the number of support vectors used to construct the regression function. The bigger ϵ , the fewer support vectors are selected, the "flatter" estimates.

It is associated with the choice of the loss function (ϵ -insensitive loss function, quadratic loss function or Huber loss function, etc.)

- Parameter $C \left(\frac{1}{\lambda} \right)$ determines the trade off between the model complexity (flatness) and the degree to which deviations larger than ϵ are tolerated.

It is interpreted as a traditional regularization parameter that can be estimated by cross-validation for example

When the data is non-linear, use a map φ to transform the data into a higher dimensional feature space to make it possible to perform the linear regression.



Non-linear Regression and Kernels (Cont)

Suppose we consider approximation of the regression function in term of a set of basis function $\{h_m(x)\}$, $m = 1, 2, \dots, M$:

$$f(x) = \sum_{m=1} \beta_m h_m(x) + \beta_0$$

To estimate β and β_0 , minimize

$$H(\beta, \beta_0) = \sum_{i=1}^N V(y_i - f(x_i)) + \frac{\lambda}{2} \sum \beta_m^2$$

for some general error measure $V(r)$. The solution has the form

$$\hat{f}(x) = \sum_{i=1}^N \hat{\alpha}_i K(x, x_i)$$

with $K(x, x') = \sum_{m=1}^M h_m(x) h_m(x')$